

Presentation 3

What did each of the team members do?

Siddharth (math): Worked on task 1, observing the role of slice attention and whether it should be removed, and task 3, making a recommendation for treatment of resolution.

Dustin and Eric (coding): Implemented the mandatory tasks and optional task 3.

Andrew and Kai (hyperparameter tuning): Conducted Optuna Bayesian optimization on the six specified hyperparameters. Executed 3 separate Optuna runs, with different run configurations, and explored slightly different hyperparameter search spaces. I adjusted the number of epochs and added early stopping where necessary. After hyperparameter tuning runs, the best performing hyperparameters were inserted into training configuration for the complete model, and the model was trained on the full dataset. Model performance was evaluated using test accuracy, loss curves, and confusion matrices to compare how different hyperparameter configurations affected generalization.

How does it help the project?

The math recommendations will help in future adjustments of the model. Specifically, we have a recommendation to keep slice attention to prevent losing useful information as our slice attention tends to match scenario 2/3 from the slides (mixed/peaky). Also, we are aiming to do a grid search, independently tuning R_{max} (resolution) and P (patch size in pixels). Depending on the result, we can decide whether to use fixed with best tuned R_{max} as the fixed value or variable resolution. This is a major structural hyperparameter which will influence our model's generalization bound, entropy regularization, lesion sensitivity, and memory. Proper treatment will have a large impact on our model.

The code changes improved the accuracy of the model substantially and we are able to have a better understanding of how the model classifies by seeing the patch-level predictions.

For hyperparameter tuning, we identified which training configurations harm the model, for example: too high classifier dropout, too low patch projector dropout, too high benign class weight. The process established a "stable learning-rate region", and gave us insight into which hyperparameter ranges will be more effective for the next presentation.

Issues faced that haven't been resolved:

We still need to understand why case 86 is misclassified (attention issue vs. other issue?)

Proposed next steps:

Keep slice attention. Grid search to independently tune R_{max} and P , scale λ accordingly (divide by $\log(R^2/P^2)$). Depending on the result, we can decide whether to use fixed with best

tuned $R_{\max}$ as fixed value or variable resolution. From our last presentation: Remove entropy regularization from slice and stain levels.

Deeper dive into attention analysis combined with patch-analysis to understand what the model focuses on.

For hyperparameter tuning, starting with a broad search and more targeted hyperparameter search ranges will streamline the process and our results.